Aleph Psychometrics Text-to-Big5 Model Report

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Abstract

We investigate how well fine-tuned encoder models can predict a person's Big 5 personality traits from their writing, and how the choice of training data affects this across different contexts. We fine-tuned ModernBERT-large with a five-output regression head on several data mixtures, combining the Essays dataset, the PAN 2015 Author Profiling dataset (tweets), the PANDORA dataset (Reddit comments), and a synthetic dataset of Reddit-style comments that we generated with Claude models. We evaluated each model on the PAN 2015 English test set and on the Aleph Big 5 Reddit dataset, a new test set of 142 Reddit users who took our personality quiz. Our best model (FullP, trained on all four sources) reaches a mean Pearson correlation of .398 on PAN 2015 and .141 on the Aleph Reddit test set (90% CI [.087, .194]), showing that personality signal is much easier to recover in-domain than in an unseen context. We also find that removing the Essays data gives the worst performance on Reddit, and that a second stage of fine-tuning on real Reddit users, evaluated with cross-validation, does not improve performance.

1 Introduction

A human's personality affects every facet of their life, but it does so to differing extents. Not every context allows for people's natural tendencies and preferences to play a major part. For example, one would expect that personality would play a much larger part in a painter's process and artwork when compared to the process and output of a data entry clerk. The difference between the two situations seems to be that the former has a lot fewer restrictions than the latter, and there is a heavier expectation of what a successful output looks like in the latter as well. Even within a single domain, there can exist varying amounts of constraints and expectations which affect how much of someone's personality the process and output can allow for. Writing, which is one of the most important forms of human communication, is one such example. Writing can range anywhere from heavily constrained to fully unconstrained. Academic writing, for example, expects papers to be structured in a standardized format and to be written in a dry and formal style. On the other hand, informal internet forum posts on sites like Reddit and Twitter have a significantly lower level of expectations and constraints, allowing users to choose which posts and communities they engage with, as well as allowing them to have a high degree of control over the content and style of their posts. This liberty allows for users to express themselves freely, and it also allows for personality to be more easily measurable. This makes these contexts ideal for NLP systems that aim to predict personality given text as an input. However, given that education, culture, and context also play a very important role in writing style, we cannot attribute writing style to personality exclusively. Regardless, there exists a correlation between personality and writing style, even if it is not particularly strong (Gjurković et al., 2021).

The method of measuring personality that our experiments will be focusing on is the Big 5 Personality Trait Model, which is composed of five main domains: openness, conscientiousness, extraversion, agreeableness, and neuroticism. The rationale for picking this over alternatives such as MBTI or the Enneagram is that, according to the literature, it has a high degree of scientific reliability and validity (John et al., 2008; McCrae and Costa, 1997). The challenge is that it is not as popular as MBTI or the Enneagram, which makes labeled data for it more scarce. Part of the project centers around navigating these obstacles and determining which strategies are the most successful at addressing these issues.

We are interested in investigating a few things. We want to explore to what extent recent encoder models are capable of identifying the personality signal found in different contexts, whether a single model can have strong performance across contexts, and whether models can generalize to unseen contexts. We also want to explore which data mixture strategies are best for the aforementioned questions, and the role that synthetic data can play in the development of these models.

2 Methodology and Results

We utilized several datasets for this project, some of which were publicly available, and others that were generated or collected for the express purpose of developing and testing our models. The public datasets we used include the Essays dataset (Pennebaker and King, 1999)(which consists of stream-of-consciousness essays alongside a binary label for each Big 5 trait of the author), the English part of the PAN 2015 Author Profiling dataset (Rangel Pardo et al., 2015) (which contains tweets alongside the Big 5 traits of the author), and the PANDORA dataset (Gjurković et al., 2021) (which contains Reddit comments and the Big 5 traits of the author).

We created two datasets for this project, and these were the Aleph Big 5 Synthetic dataset and the Aleph Big 5 Reddit dataset. The Synthetic dataset was created by first generating around 1,000 personas, each with an age, predetermined Big 5 scores, and interests; we passed 600 personas to Claude Haiku 4.5 and 400 personas to Opus 5.5. We asked these models to generate text in the Reddit comment style conditioned on the Big 5 scores, interests, and age of each persona. We then divided the output for each persona into chunks (although for most personas, the entire output fit inside a single chunk). For the Aleph Big 5 Reddit dataset, we simply collected the Big 5 scores and Reddit username combos for the respondents of the quiz who agreed to provide this information (of the 682 quiz takers, we had 142 users voluntarily provide a valid Reddit username). We then fetched the posts of these users and divided them into chunks.

For testing our model, we used two test sets: the PAN 2015 Author Profiling English test set and our own Aleph Big 5 Reddit dataset. These two test sets were selected to evaluate how well our model would perform on different domains and to allow us to evaluate the cross-domain transfer of our model.

We created different models using different data mixtures in order to determine which data mixtures worked best for our purposes. For all models, we used ModernBERT-large as a base, and we added a regression head of five outputs (one per Big 5 trait) using the mean-pooled token embeddings of the encoder. We then fine-tuned the full model on each data mixture (keeping the hyperparameters constant) so that we could isolate the effect of each data mixture on the model's performance. We evaluated each model on the same two test sets we outlined earlier. We used the model trained exclusively on Essays as a baseline due to its simplicity.

Optimizer	AdamW, weight decay 0.01
Learning rate	2×10^{-5}; 10% warmup, then linear decay
Effective batch size	16 (4×4 accumulation)
Epochs	8
Checkpoint	lowest dev loss (4 evals/epoch)
Max sequence length	4096 tokens
Loss	MSE, labels scaled to $[0, 1]$
Precision	bf16, gradient checkpointing
Seed	0 (one run per model)

Table 1: Fine-tuning hyperparameters, fixed before any experiment and identical across all data mixtures.

We tested the following data mixtures: Essays (only includes the Essays dataset), NoEssays (only includes PAN2015 and Synthetic), Full (includes Essays, PAN2015, and Synthetic), FullP (includes Essays, PAN2015, Synthetic, and Pandora).

We also tested whether a second stage of fine-tuning on real Reddit user data would improve the model's performance on this domain. We used the FullP model as a base. The only Big 5 data left to use was the Aleph Reddit test set itself, and since we wanted to avoid training on the testing set, we used 5-fold cross-validation, and fine-tuned on 4 folds (using a smaller learning rate for 2 epochs) and then evaluated this model on the 5th fold. We did this 5 times so that each fold was evaluated by a model that did not include it in the

training process. By combining these evaluations, we obtained evaluation results on the test set for this model, without data leakage.

3 Discussion and Limitations

The results indicate several things. Firstly, not every Big 5 trait is equally easy to identify via text. Across the Aleph Reddit test set, agreeableness (A) is consistently the trait with the highest Pearson correlation between predicted and actual scores for all our models, and by a substantial margin. The second-highest trait would be openness (O). The hardest trait to identify in the Aleph Reddit test set is extraversion (E). The story is slightly different for the PAN2015 test set. The trait with the highest correlation between predicted and actual scores is openness (O), with agreeableness (A) second. The main difference between the two test sets is the variance of the average correlation scores, with the Aleph Reddit test set showing significant variance, whereas the average correlation scores for PAN15 tend to be significantly closer together.

Another thing to note is that the Essays model performs pretty poorly for both test sets, compared to most alternatives. Another interesting finding is that the NoEssays model has the best performance out of all of our models on the PAN15 test set, and this makes sense because, out of all models, it has the highest proportion of PAN15 data in its training set, which highlights the importance of in-domain data. The Full model is an improvement over both the Essays and NoEssays models for the Reddit test set, while still remaining fairly competitive with the NoEssays model for the PAN15 test set. The FullP model is an improvement over the Full model on both test sets, which means that adding PANDORA's dataset into the mix improves performance across both of these domains. Our stage-2 CV experiment using FullP as a base was the second-worst-performing model on the Reddit test set, which gives us reason to believe either that staging is unhelpful in this case or that our models are incapable of extracting valuable signal from the Reddit test set. In any case, FullP is the strongest model according to our results, being the best performer on the Reddit test set and the second-best performer on the PAN15 test set. This means that one model can perform reasonably well across different domains.

The project had some important limitations. Firstly, the composition of the dev set, coupled with the role that the dev set played in model selection, may have led to suboptimal models being selected for each mixture. The data composition of the dev set for each model trained was roughly proportional to the training data. Additionally, during training, the final model for each data mixture was the model with the best dev set performance, so, given that essays made up a significant portion of most models, we were picking the model based largely on its performance on a domain that is different from our target (informal social media posts), which is not ideal. Another limitation is that the size of the test set is still small, limiting the usefulness of our confidence intervals and making comparisons between models more challenging. Additionally, we did not test different seeds due to computational constraints.

References

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Model	O	C	E	A	N	Mean
Aleph Reddit test set ($n = 142$ users)						
Essays	.082 [−.033, .198]	.037 [−.077, .159]	.155 [.022, .274]	.178 [.063, .286]	.091 [−.045, .239]	.109 [.065, .153]
NoEssays	.107 [.002, .206]	−.020 [−.171, .130]	−.037 [−.175, .089]	.172 [.048, .295]	.094 [−.031, .228]	.063 [.008, .120]
Full	.108 [−.012, .226]	.089 [−.042, .216]	.072 [−.064, .192]	.227 [.114, .347]	.105 [−.040, .255]	.120 [.068, .172]
FullP	.160 [.041, .278]	.129 [−.011, .259]	.052 [−.084, .180]	.275 [.152, .396]	.086 [−.048, .214]	.141 [.087, .194]
FullP → stage 2 (CV)	.101 [−.011, .202]	.088 [−.049, .211]	−.032 [−.171, .098]	.232 [.106, .360]	.055 [−.082, .207]	.089 [.040, .138]
Average	.112	.065	.042	.217	.086	.104
PAN15 English test set ($n = 142$ users)						
Essays	.132 [−.007, .262]	.133 [.004, .268]	−.072 [−.187, .053]	.263 [.143, .381]	−.044 [−.186, .098]	.082 [.023, .145]
NoEssays	.528 [.424, .622]	.316 [.205, .426]	.473 [.374, .569]	.421 [.330, .508]	.500 [.379, .608]	.448 [.398, .499]
Full	.418 [.314, .520]	.386 [.272, .498]	.393 [.289, .497]	.322 [.225, .418]	.377 [.269, .476]	.379 [.328, .428]
FullP	.409 [.292, .514]	.402 [.265, .528]	.398 [.297, .494]	.323 [.224, .415]	.458 [.350, .559]	.398 [.345, .450]
Average	.372	.309	.298	.332	.323	.327

Table 2: Pearson r between predicted and questionnaire scores, with 90% bootstrap confidence intervals (2,000 resamples over users). CV rows are pooled out-of-fold predictions from 5-fold cross-validation over the Aleph users, so each user is predicted by a model that never saw them; they have no PAN15 result because each fold model predicted the whole PAN15 test set.